Automatic Extraction of Roads From Multisource Geospatial Data Using Fusion Attention Network and Regularization Algorithm

Zejiao Wang, Longgang Xiang, Zhongyu Liu, and Zhengxiang Wang

***Abstract*— Automatic extraction of roads from remote sensing imagery has numerous applications, such as urban planning and navigation. However, the quality of images is limited, and most existing road extraction methods suffer from discontinuity. Additionally, there remains a gap between pixel-based road segmentation and road vectorization. To address these challenges, we introduce a method that utilizes a multistage feature fusion attention network (MFFANet) and regularization algorithm to extract road surfaces, centerlines, edges, and intersections. MFFANet comprises three components. The complementary feature embedding module (CFEM) adaptively encodes remote sensing images, vehicle trajectories, and OpenStreetMap (OSM) points to capture specific modal features. The multistage feature fusion module (MFFM) is proposed to improve the completeness and connectivity of road extraction by integrating multisource geospatial features. The multilevel mask generation module (MMGM) enhances road segmentation results through a weighting mechanism that can simultaneously predict local road segments, road sections, and road network masks. Additionally, a novel joint loss function is introduced to balance local and global optimization. In the regularization stage, fused hierarchical masks generate road segments, with a skeleton refinement for centerlines and widths, followed by smooth segment reconstruction and extraction of edges and intersections. Experiments on different datasets demonstrate that our designed fusion attention network outperforms the latest road segmentation models; our regularization algorithm shows strong robustness and the comprehensive metrics of vectorized road line extraction exceeds 70%.**

***Index Terms*— Fusion attention network, multilevel mask generation, multisource data, regularization algorithm, road centerline vectorization, road extraction, road surface inference.**

I. Introduction

AUTOMATIC road extraction is a fundamental and important problem in the field of geoinformation science, with numerous applications including navigation, updating geoinformation databases, emergency support scenarios, and autonomous driving. Additionally, road extraction provides contextual information which is beneficial for other related tasks, such as urban planning and vehicle detection. The road extraction methods can be primarily divided into traditional and deep learning-based methods [1]. Traditional machine learning methods have attempted to recognize roads using textures, lines, shadows, and other more complex and empirically designed features. However, these approaches suffer from low automation and poor generalization capabilities. Recently, the development of deep learning has driven a new wave of research in automated road extraction and achieved better performance compared with traditional methods, mainly due to the powerful self-learning and feature expression capabilities. However, many critical issues still exist, such as the limited quality of a single data source of remote sensing image, which often results in discontinuities in existing road extraction methods due to occlusion. Additionally, a gap between pixel-based road segmentation and road vectorization remains. The research on road extraction can be divided into road surface segmentation, centerline extraction, edge line extraction, and intersection extraction. In previous works [2], many methods have been developed to extract road surfaces using single data source of remote sensing images. In this study, we focus on integrating the high-quality advantages of multisource geospatial data to extract road surfaces, road centerlines, road edges, and intersections simultaneously, which is a challenging task.

For road surface extraction, researchers have proposed a wide range of methods. Das et al. [3] proposed a multistage framework that includes four support vector machine (SVM) and a series of postprocessing to extract roads from satellite images. Wang et al. [4] utilized spectral and texture features for road identification. The emergence of fully convolutional networks [5], as the first end-to-end network designed for pixel-level prediction, significantly improved road extraction performance. Guo et al. [6] proposed a dual-path deep learning network capable of accurately extracting narrow roads. Xu et al. [7] proposed a dual-branch partial-to-complete road extraction network based on satellite images and partial road maps. Qiu et al. [8] proposed a cross-modal information propagation mechanism that utilizes scaling factors and sparse constraints to distinguish the redundant information of specific modalities. Meng et al. [9] proposed a multilayer attention fusion network that integrates multiple layers of transformer [10] features, yielding more comprehensive semantic information. RoadCT [11] combines the advantages of convolution and transformer networks to enhance road extraction performance. Ground object occlusion and capturing global road features are typical challenges constraining

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road segmentation. These issues have been actively researched and remain in the development stage.

For road centerline extraction, most works are implemented by two steps, where the road surface is first detected and then refined to extract the centerline. For example, Huang and Zhang [12] proposed a method based on multiscale structural features and SVM to extract road centerlines from high-resolution images. Lourenço et al. [13] proposed a method, which first predicts the rural road surfaces and then uses morphological techniques to extract the road centerlines. With the development of deep learning, many end-to-end centerline extraction methods have emerged. Lin et al. [14] proposed a dual-task-driven deep learning framework that can simultaneously perform road surface segmentation and centerline extraction tasks. Yang et al. [15] constructed the road centerline topological structure with graph neural networks. TopoRoad [16] is an end-to-end road graph extraction method that leverages a road topology map to guide the learning of vectorized representations of road maps. More recently, some transformer-based studies have been introduced. Xu et al. [17] proposed a road extraction method based on transformer and imitation learning, which can directly obtain road networks. Road network graph detection++ (RNGDet++) [18] builds upon [17] with improvements, designing a multiscale feature fusion module and adding an instance segmentation head for supervised training. Hetang et al. [19] proposed SAM-Road, which leverages the inherent strengths of segment anything model (SAM) [20] and designs a lightweight transformer-based graph neural network to predict the centerlines of roads. Liao et al. [21] proposed MapTR, which can generate vectorized high-definition maps online. However, issues of vertex redundancy and edge topological connection errors still exist.

Extracting edge lines of roads from remote sensing images has been rarely studied. Extracting road edges accurately in the face of complex road environment is challenging. Suleymanoglu et al. [22] proposed a novel methodology based on a machine learning strategy, which enables high automation and extracts road boundaries from mobile light detection and ranging system (MLS) and mobile mapping system (MMS) point cloud data. In summary, previous studies have struggled to achieve accurate pixelwise road segmentation, making it challenging to extract road edge vector lines through postprocessing. Xu et al. [23] proposed an innovative solution for road edge detection based on imitation learning, which fully leverages historical information to generate road edge graphs from remote sensing images. PolyRoad [24] is an innovative polyline transformer method for topological road boundary detection, capable of parallel road boundary detection. LCDeT [25] is an efficient 3-D curb detection network integrated with a transformer, achieving end-to-end curb extraction from mobile laser scanning data. Moreover, current end-to-end methods often have topological errors and omissions.

Intersections, as crucial nodes within road networks, play a vital role in practical applications. Deng et al. [26] proposed a trajectory gridding strategy based on hierarchical partitioning to address the spatial distribution heterogeneity of trajectory density, along with a comprehensive method for road intersection extraction. Yang et al. [27] obtained detailed 3-D structures of overpasses through crowdsourced trajectories, then identified grade-separated road intersections. He et al. [28] proposed a method that combines object detection and iterative node search to detect road intersections. Li et al. [29] developed two intersection detection networks, and achieved promising results using LiDAR data. Currently, researchers have investigated various intersection detection methods using multiple data sources. However, the automatic extraction of road intersections still encounters several challenges, including the usage of single data source and issues related to data quality.

According to the analysis above, it can be inferred that there is considerable research focused on the automatic extraction of various types of roads. Most existing road extraction methods suffer from discontinuity due to the limited quality of remote sensing images. Additionally, the gap between pixel-based road segmentation and road vectorization still exists. To solve these problems, this study proposes a multistage feature fusion attention network (MFFANet) and a multilevel road mask fusion strategy to obtain accurate binary road maps. Simultaneously, it employs a robust and effective empirical regularization algorithm to extract road centerlines, edge lines, and road intersections ultimately. We evaluate the proposed method on two datasets, demonstrating that the designed fusion attention network outperforms the state-of-the-art road segmentation models and the comprehensive metrics of vectorized road line extraction exceeds 70%.

The main contributions of this work are summarized as follows.

1) We propose a method that utilizes a well-designed fusion attention network and robust regularization algorithm to extract road surfaces, centerlines, edges, and intersections simultaneously from multisource geospatial data.
2) A multistage feature fusion module (MFFM) is proposed to improve the completeness and connectivity of road extraction by integrating multisource geospatial features. The multilevel mask generation module (MMGM) simultaneously predicts masks of local segments, road sections, and road network through a weighting mechanism. Additionally, an effective joint loss function is designed to balance local and global optimization.
3) We propose a novel regularization algorithm to generate smooth, topologically correct road surfaces, centerlines, and edges. It denoises and fuses hierarchical masks, refines centerlines, reconstructs road segments, and uses edge detection and neighborhood search to extract edges and intersections.

The remainder of this article is organized as follows. Section II describes our methods, including the MFFANet and the empirical regularization. Section III elaborates on the experiments and results, while Sections IV and V present the discussion and conclusions, respectively.

II. Methods

The road extraction method includes pixel-based segmentation and regularization. First, we develop an MFFANet architecture to extract road pixels. The MMGM can simultaneously predict local road segments, road segments, and road

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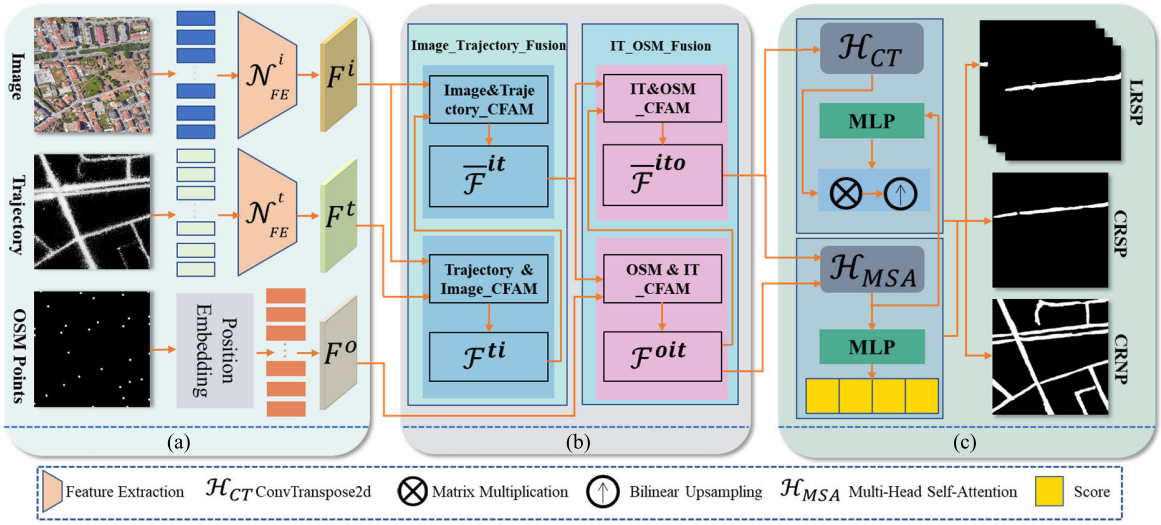


Fig. 1. MFFANet structure. (a) Complementary feature embedding. (b) Multistage feature fusion. (c) Multilevel mask generation.

network masks, ensuring the quality of regularization process. Then, the multilevel masks are processed by our regularization algorithm to produce road surfaces, centerlines, edge lines, and intersections.

A. Multistage Feature Fusion Attention Network

The structure of the MFFANet is shown in Fig. 1. To extract road pixels efficiently and completely, the model focuses on complementary feature fusion across multisource data, in contrast to previous works that emphasize fusing multiscale features [30], [31]. Compared with single data, multisource data can describe the characteristics of roads from various perspectives, and information missed in one source can be supplemented by another. We designed the complementary feature embedding module (CFEM) and MFFM. CFEM extracts specific modal feature representations from multisource data, while the MFFM achieves deep fusion of complementary features from multiple sources. MFFM weakens features that interfere with road detection from single data source and fuses effective road features dispersed across different data sources. To improve the quality of road pixel extraction and regularization, in contrast to the previous single mask prediction [32], [33], we designed a MMGM, which can simultaneously predict masks of local segments, road sections, and road networks. The multilevel results consider both local and global road pixel prediction, making road pixel extraction more reliable.

1) Complementary Feature Embedding: Extracting roads from a single image data source presents numerous challenges mainly due to the occlusion caused by trees and buildings, which often result in fragmented and poorly connected road segments in road networks. Compared with single data source, multisource data can describe the characteristics of roads from different perspectives, and information missed in one source can be complemented in another. Trajectory data has the advantage of generating continuous spatial topology, while OpenStreetMap (OSM) points have good road positioning, both of which are lacking in images. We considered the characteristics of remote sensing images, vehicle trajectories, and OSM points and designed a CFEM, as shown in Fig. 1(a). This module embeds modality-specific features into the network to achieve optimal complementary fusion of road features. The SAM, as a foundational visual segmentation model, has excellent generalization capabilities. We adopt the pretrained encoder from SAM, which is the smallest ViT-B variant, as our encoder for images and vehicle trajectories. It utilizes the ViT_Base [34] architecture, mainly composed of a patch embedding layer, 12 transformer encoder layers, and two convolutional layers. Specifically, given the image $X_{\text{image}} \in \boldsymbol{R}^{1024\times1024\times3}$ and the trajectory density map $X_{\text{trajectory}} \in \boldsymbol{R}^{1024\times1024}$, we feed them into autoencoders to learn modality-specific representations explicitly, resulting in image features $F^i \in \boldsymbol{R}^{256\times64\times64}$ and trajectory features $F^t \in \boldsymbol{R}^{256\times64\times64}$, where 256 is the number of channels, and the width and height are both 64. By optimizing with independent parameters, these features can effectively preserve the specific information of each modality, thus allowing for further mutual refinement.

The OSM data consists of a series of discrete points, which should be located on roads in theory. In this study, to enhance road feature perception, we map OSM points to position encodings and embed them into the network as prior guidance information of local road. Point position encoding is achieved using a straightforward Fourier feature mapping mechanism [35], as shown in the following equation:

$$\mathcal{F}(x) = \text{Concatenate}\left(\cos\left(f(x)k^{\mathrm{T}}\right) + \sin\left(f(x)k^{\mathrm{T}}\right)\right) \quad (1)$$

where x denotes the OSM point vector, and k represents the Gaussian matrix. $f(\cdot)$ denotes the mapping of points from a low-dimensional to high-dimensional representation, and Concatenate$(\cdot)$ denotes tensor concatenation operation. After applying (1), the points are represented in a 256-D positional embedding format F^o.

2) Multistage Feature Fusion: After obtaining the complementary feature embeddings, we employ an MFFM to refine these features mutually, as shown in Fig. 1(b). By absorbing

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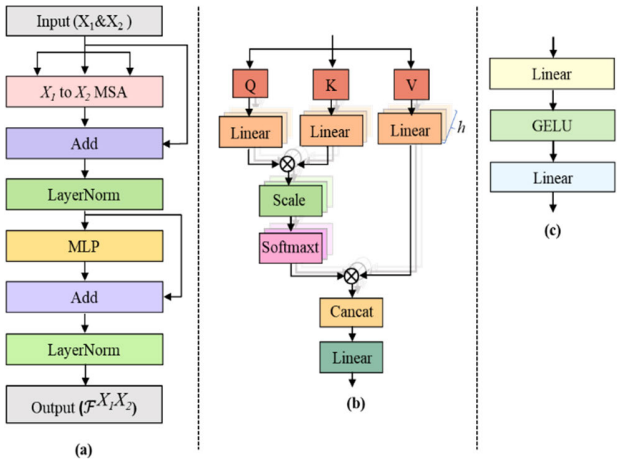


Fig. 2. Flowchart of (a) CFAM, (b) MSA, and (c) MLP.

the complementary information from other modalities, each unimodal feature becomes more reasonable and robust. MFFM consists of multiple cross-modal fusion attention modules (CFAMs), as illustrated in Fig. 2(a). In contrast to previous works [36], [37] that relied on elementwise multiplication or concatenation strategies for feature fusion, we employ a fusion attention mechanism. This approach mutually enhances the features through a message-passing mechanism to capture the complementary information from multiple modalities fully.

Specifically, as illustrated in Fig. 1(b), we implement the fusion in two stages. In the first fusion stage, we achieve a deep fusion of image features and trajectory features. The module first performs CFAM between F^t and F^i. We reshape F^t into a tensor of shape (b, 4096, 256) to serve as the query, while F^i is reshaped into a tensor of the same shape to act as the key and value, where b represents the batch size. Then, a multihead self-attention (MSA) layer processes these tensors to generate A. The procedure is shown in Fig. 2(b), and the key calculation of this process is expressed as follows:

$$\text{Attention}\left(F^t, F^i, F^i\right) = \text{softmax}\left(\frac{F^t\left(F^i\right)^{\mathrm{T}}}{\sqrt{d_{F^i}}}\right)F^i \qquad (2)$$

where d_{F^i} denotes the dimensionality of F^i. Subsequently, A is added to F^t and layer normalization is applied to obtain A'. We use a multilayer perceptron (MLP) that consists of two linear layers and an activation layer to better capture the potential features of the road, as shown in Fig. 2(c). Then the output is processed by a residual shortcut and layer normalization to generate $\mathcal{F}^{ti}$. This process can use the trajectory features to enhance the high-value features of image, which is beneficial for discovering occluded roads. At this point, trajectory continuity features are fused into the image. To achieve bidirectional flow enhancement for trajectory and image information, we feed F^i and the fused $\mathcal{F}^{ti}$ into the CFAM module to generate $\overline{\mathcal{F}}^{it}$, where F^i is used as the query, and $\mathcal{F}^{ti}$ serves as the key and value. This process further leverages visual information to eliminate some useless and noisy road features effectively, uncovering more valuable features in trajectory and image, thus ensuring the completeness of road extraction.

In the second fusion stage, we achieve the fusion of OSM points with image and trajectory data. The model first applies CFAM between F^o and $\overline{\mathcal{F}}^{it}$, where F^o is used as the query, and $\overline{\mathcal{F}}^{it}$ serves as the key and value. We feed F^o and $\overline{\mathcal{F}}^{it}$ into MSA to obtain B. Then, we add B to F^o and apply layer normalization to obtain B'. Finally, B' passes through an MLP, a residual shortcut, and layer normalization to generate $\mathcal{F}^{oit}$. This process integrates the prior knowledge from the OSM points, enabling the model to focus more on the road areas in the image and trajectory, which is crucial to distinguishing some illusory nonroad areas. Furthermore, to achieve bidirectional enhancement of image, trajectory, and OSM points information, we feed $\overline{\mathcal{F}}^{it}$ and $\mathcal{F}^{oit}$ into the CFAM module to generate $\overline{\mathcal{F}}^{ito}$, where $\overline{\mathcal{F}}^{it}$ serves as the query and $\mathcal{F}^{oit}$ serves as the key and value. This further fuses the complementary features of the roads among image, trajectory, and OSM points, which then help in reconstructing roads obscured by strong noise, and is vital for ensuring the integrity of the roads.

3) Multilevel Mask Generation: In the past, road surface extraction algorithms only predicted road network masks, ignoring the prediction of local segments and segment masks. In this study, we designed the MMGM to extract masks of local segments, segments, and road network simultaneously. The multilevel results provide a comprehensive description of the road surface from local and global perspectives, enhancing the effectiveness of road surface extraction. In the MFFM stage, information from various modalities is fused together, effectively capturing the deep semantic features of the road. However, to highlight the importance of a certain modal feature at specific locations for road extraction, we designed a sophisticated weighting mechanism that amplifies high-value features and reduces noise interference. In our design, local segments serve as the primitives for hierarchical mask output. To ensure accuracy, we performed four local segment predictions for each position point while simultaneously predicting the correctness score of each result. The final output is the optimal prediction, selected on the basis of the highest score.

As shown in Fig. 1(c), in the MMGM, we obtain the fused feature maps from MFFM, which can be expressed as $\overline{\mathcal{F}}^{ito} \in \boldsymbol{R}^{b\times 4096\times 256}$ and $F^{oit} \in \boldsymbol{R}^{b\times 4\times 256}$. $\overline{\mathcal{F}}^{ito}$ and $\mathcal{F}^{oit}$ are first passed through the MSA layer to obtain the weight tokens and score tokens. Then, they are fed into the MLP layer separately to obtain weights W and scores S. Meanwhile, the reshaped $\overline{\mathcal{F}}^{ito}$ is processed by two deconvolution layers to obtain F_M, where the number of channels decreases from 256 to 32, and the resolution increases from 64 × 64 to 256 × 256. We further perform a matric multiplication between W and the reshaped F_M and reshape the result to $\boldsymbol{R}^{b\times 4\times 256\times 256}$. After applying bilinear upsampling, we obtain four predictions of the local segment mask for the same local point. Finally, the mask with the highest score is selected as the local road segment prediction (LRSP). After predicting all local segments on one road section, we fuse all LRSPs to obtain complete road segment prediction (CRSP). Once the inference on one image is completed, we combine all CRSPs to obtain the complete road network prediction (CRNP). This study is the first to extract local road segments, road segments, and road network masks simultaneously, which is crucial for obtaining complete

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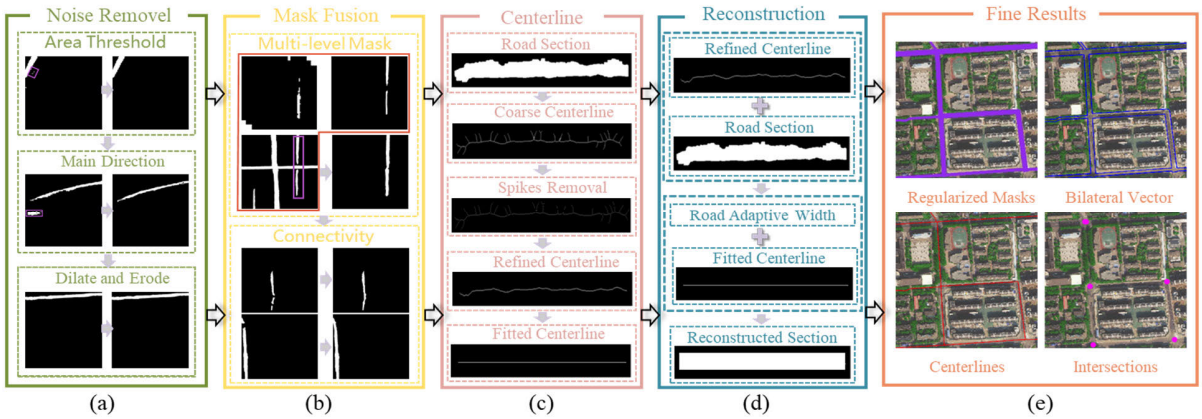


Fig. 3. Overall flowchart of regularization algorithm. (a) Noise removal. (b) Multilevel mask fusion. (c) Centerline extraction. (d) Road reconstruction. (e) Fine results.

and continuous roads. Additionally, it ensures the quality of our empirical regularization.

4) Loss Function: Our MMGM stage has three kinds of prediction: LRSP, CRSP, and CRNP, and our total loss is related to these three components. Dice loss [38] is commonly used to address class imbalance, while focal loss [39] enhances the model's ability to focus on hard-to-classify samples in imbalanced datasets. In contrast, mean squared error (MSE) loss provides stability when predicting continuous values. Road extraction is characterized by class imbalance and the challenge of pixel recognition. To address this, we calculate the loss for each component using a linear combination of focal loss, dice loss, and MSE loss, ensuring improved performance in handling these difficulties. The mathematical formula can be expressed as follows:

$$L_{\text{LRSP}} = \beta L_{\text{Focal}}^{\text{LRSP}} + \eta L_{\text{Dice}}^{\text{LRSP}} + \zeta L_{\text{MSE}}^{\text{LRSP}} \tag{3}$$

$$L_{\text{CRSP}} = \beta L_{\text{Focal}}^{\text{CRSP}} + \eta L_{\text{Dice}}^{\text{CRSP}} + \zeta L_{\text{MSE}}^{\text{CRSP}} \tag{4}$$

$$L_{\text{CRNP}} = \beta L_{\text{Focal}}^{\text{CRNP}} + \eta L_{\text{Dice}}^{\text{CRNP}} + \zeta L_{\text{MSE}}^{\text{CRNP}} \tag{5}$$

where β, η, and ζ are the weight hyperparameters. L_{Focal}, L_{Dice}, and L_{MSE} are the focal loss, dice loss, and MSE loss, respectively. The formulas for these losses are computed as follows:

$$L_{\text{BCE}} = -\frac{1}{N}\left(\sum_{Y_i=1}^{m} Y_i \log\left(\hat{Y}_i\right) + \sum_{Y_i=0}^{n} (1 - Y_i)\log(1 - \hat{Y}_i)\right) \tag{6}$$

$$L_{\text{Focal}} = -\alpha(1 - f(L_{\text{BCE}}))^{\gamma} L_{\text{BCE}} \tag{7}$$

$$L_{\text{Dice}} = 1 - \frac{2\left|\hat{Y} \cap Y\right| + \epsilon}{\left|\hat{Y}\right| + |Y| + \epsilon} \tag{8}$$

$$L_{\text{MSE}} = \frac{\sum_{i=1}^{n}\left(\hat{Y} - Y\right)^2}{n} \tag{9}$$

where $\hat{Y}$ is the predicted road mask and Y is the ground truth. The formula for the total loss is expressed as follows:

$$L_{\text{total}} = L_{\text{LRSP}} + L_{\text{CRSP}} + L_{\text{CRNP}} \tag{10}$$

where L_{total} considers both local and global optimality, enhancing the completeness of road extraction and remarkably improving the continuity of the results.

B. Regularization Algorithm

Fig. 3 illustrates the pipeline of our regularization algorithm, which primarily consists of noise reduction, mask fusion, centerline extraction, and road segment reconstruction. The final output includes the regularized road surfaces, centerlines, edge lines, and intersections.

1) Noise Removal: The multilevel road mask output by the MFFANet inevitably contains noise, holes, and jagged edges. Leveraging prior knowledge of road geometric structures, we apply a refined processing to the binary road segmentation image. The key steps are as follows.

1) Employing the boundary tracking algorithm [40], we extract the boundary polygons of all continuous regions in the binary road image. We set an area threshold S; any region with an area smaller than S is considered noise and removed accordingly.
2) We utilize the main direction of the contours to remove noise blocks with areas larger than S. First, we identify the contours of all continuous regions in the binary image and calculate two key metrics for each contour: the aspect ratio of its minimum bounding rectangle and the distance between the two farthest points within the contour. On the basis of these metrics, the potential road blocks are initially selected to be retained. Then, for the first retained block, we compute its center point P_i and main direction α using the first and second moments of the image. For each of other continuous blocks, we calculate the direction angle β between its center point and P_i. If the absolute difference between α and β exceeds a threshold ε, the block is considered noise and discarded.
3) Dilation and erosion morphological operations are applied to remove the jagged effects of the binary image edges. Specifically, we use a 5×5 rectangular kernel for three dilation and erosion operations on the results obtained from step 2).

2) Mask Fusion: Inspired by the law of large numbers in probability theory, which states that as the number of independently and identically distributed random variables increases, the sample mean tends toward the expected value, we integrate

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the multilevel road masks to enhance the accuracy of road extraction. For the same road segment, given n predicted results in LRSP and CRSP, we identify all contours with the maximum area-to-perimeter ratio among the results and obtain nr_{ap}. Then, we calculate the mean of the n maximum area-to-perimeter ratio values, denoted as mr_{ap}. The predicted results with r_{ap} values exceeding mr_{ap} are combined to generate the fused mask F_s^1. We compute the minimum bounding rectangle of F_s^1 and use it as the cropping range to extract road segment information F_s^2 within the corresponding area of CRNP. F_s^1 and F_s^2 are merged to produce the multistage result F_s, and disconnection processing is applied to F_s to ensure the continuous connectivity of individual road segments. By merging all F_s within a single sample, we obtain the final fused result of the segmented road network, denoted as F.

3) Centerline Extraction: Road centerline reconstruction is an essential step in regularization algorithms, designed to obtain results close to the actual road centerline. The detailed implementation process is illustrated in Fig. 3(c).

1) By utilizing the parallel thinning algorithm with two subiterations [41], the coarse road centerline is extracted from F_s.
2) Localization and removal of spikes in the coarse centerline. All node positions in the coarse centerline are identified using the eight-neighborhood algorithm, after which the nodes are labeled, and background pixel values are assigned to their surrounding eight neighbor pixels. We discretize the coarse centerline into line segments, using nodes as breakpoints. Spikes can then be accurately located and removed on the basis of criteria, such as the aspect ratio of the corresponding minimum bounding rectangles, the principal direction, node labeling, and the number of discrete centerline segments.
3) After removing the spikes and merging the remaining line segments, we apply a scalable border search method to eliminate any remaining spikes. This process ultimately yields the refined centerline.
4) The refined centerline contains numerous redundant points. We utilize the staged Douglas-Peucker (DP) algorithm [42] to restore the straight and smooth characteristics of the road and obtain the fitted centerline.
5) Some topological issues, such as disconnection and overconnection, occur when the centerlines belonging to the same sample are combined. We rectify them on the basis of road geometric rules.

4) Road Reconstruction: We interpolate the road centerline that has been topologically corrected and select N interpolation points. The perpendicular distances are calculated between these N interpolation points and the corresponding road segment outlines. Statistical analysis is conducted on these N distances, selecting the distance values between the lower quartile (Q_1) and upper quartile (Q_3) of the boxplot. Then, we average these distance values to obtain the width of the corresponding road segment. By using this width and the topologically corrected centerline, we generate the reconstructed road mask segments.

5) Fine Results: By merging the topologically corrected road centerlines, we obtain the road centerline network. Similarly merging the reconstructed road masks, we obtain the regularized road surface network. Edge detection is performed on the regularized mask using the Laplacian [43] operator as follows, yielding the road edge lines, where $f(x, y)$ is the grayscale value function of the image:

$$\nabla^2 f(x, y) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}. \tag{11}$$

Intersection extraction begins with an eight-neighborhood search on the centerline network to identify initial intersection points. Then, points within the same intersection range are clustered using the Density-based spatial clustering of applications with noise (DBSCAN) algorithm [44]. Moreover, the average value of each cluster is calculated to determine the final intersection points.

III. Experiment and Results

A. Datasets

To evaluate the performance of the proposed method in this study for road network extraction from multisource geospatial data, we specifically selected the open-source dataset BJRoad [45], which is known for its severe tree and building occlusions and other spectral interferences. Additionally, to validate further the effectiveness of our method for automatically extracting road networks, we developed our own dataset, PWRoad.

The BJRoad dataset, which is publicly available, was captured in Beijing, China. This benchmark comprises 350 high-resolution remote sensing images covering an area of approximately 100 km^2, along with approximately 50 million trajectory records from about 28 000 vehicles. Each remote sensing image has a size of 1024 × 1024 pixels and a resolution of 0.5 m. To fulfill our input requirements, we supplemented the dataset with the corresponding OSM data from the official website. Finally, the dataset is divided into three parts: 80% of the samples are designated for training, 10% for validation, and the remaining 10% for testing.

The PWRoad is a self-constructed dataset, which is captured from Porto and Wuhan cities. We collected satellite images from Google Earth [46] with 0.5 m/pixel resolution and downloaded OSM data from the official website. Additionally, we gathered trajectory data for both cities: the Wuhan trajectory data contains continuous taxi trajectories over ten days, while the Porto data has over one year. To meet our network training requirements, we accurately delineated road mask labels. The dataset contains 762 samples of size 1024 × 1024 pixels, with 80% used for training, 10% for validation, and 10% for testing.

In our task, all satellite images are cropped into 1024 × 1024 tiles, and the corresponding bounding box coordinates $[\text{lon}_l, \text{lat}_l] * [\text{lon}_u, \text{lat}_u]$ are recorded, where l and u represent the lower-left and upper-right positions, respectively. For the trajectory data, we filter sample trajectories based on these bounding boxes. Subsequently, we create trajectory density grayscale images, where each pixel value indicates the number of samples projected onto it. Inspired by the advantages of kernel density estimation [47] in trajectory data processing, we normalize the trajectory density maps using a logarithmic

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function and smooth them with a 3 × 3 Gaussian kernel filter to obtain the final trajectory samples.

B. Evaluation Metrics

1) Road Segmentation Metrics: Precision (P_S), recall (R_S), $F1$ score ($F1_S$), and intersection over union (IoU_S) are metrics commonly used for evaluating semantic segmentation. P_S represents the fraction of predicted road pixels that are actual road pixels, while R_S measures the fraction of actual road pixels that are correctly predicted. $F1_S$ and IoU_S are comprehensive metrics that provide a balance between precision and recall. These pixel-level evaluation metrics are defined as follows:

$$P_S = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{12}$$

$$R_S = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{13}$$

$$F1_S = \frac{2 * P_S * R_S}{P_S + R_S} \tag{14}$$

$$\mathrm{IoU}_S = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}+\mathrm{FP}} \tag{15}$$

where TP, FP, and FN represent the true positive, false positive, and false negative, respectively.

2) Road Centerline and Edge Line Metrics: We use the general GEO evaluation method (GEO method) [48] to evaluate road centerlines and edge lines. We resample both the predicted road network and the ground-truth road network globally at eight-pixel (4 m) intervals. A weighted maximum matching algorithm is used for matching, with an eight-pixel threshold for point matching and a 30° threshold for matching direction. The sampling points designated for comparison with the ground-truth road network are called "marble," while the sampling points from the ground-truth road network are called "hole." For the GEO evaluation of the road network, a global one-to-one matching approach is applied. If a "marble" has a "hole" within the matching threshold, it is termed Matched_Marbles; if a "marble" does not have a "hole" within the matching threshold, it is called Spurious_Marbles. After matching, any "holes" that do not have corresponding "marbles" are referred to as Empty_Holes, while a "hole" that contains a "marble" is called a Matched_Hole. By counting the quantities of these values, precision (P_L), recall (R_L), and $F1$ ($F1_L$) metrics can be calculated as follows:

$$P_L = \frac{\text{Matched_Marbles}}{\text{All_Marbles}} \tag{16}$$

$$R_L = \frac{\text{Matched_Holes}}{\text{All_Holes}} \tag{17}$$

$$F1_L = \frac{2 * P_L * R_L}{P_L + R_L} \tag{18}$$

where P_L and R_L are commonly used metrics to evaluate the performance of graphic sampling methods. In this context, P_L is defined as the ratio of matched samples to the total number of samples on the predicted road map, while R_L is the ratio of matched samples to the total number of samples on the ground-truth road map. P_L effectively measures the accuracy of the predictions, while R_L assesses the coverage and completeness of the ground-truth map. These metrics are often combined into $F1_L$, which is the harmonic mean of P_L and R_L, providing a more comprehensive evaluation of the prediction results.

3) Intersection Metrics: Precision (P_I), recall (R_I), and $F1$ score ($F1_I$) are used to evaluate the effectiveness of intersection detection, and the formulas are as follows:

$$P_I = \frac{p\mathrm{TP}}{p\mathrm{TP}+\mathrm{FP}} \tag{19}$$

$$R_I = \frac{t\mathrm{TP}}{t\mathrm{TP}+\mathrm{FN}} \tag{20}$$

$$F1_I = \frac{2 * P_I * R_I}{P_I + R_I}. \tag{21}$$

True intersection points are used as the center to create a buffer with a radius of R. Here, pTP denotes the number of predicted intersection points within the buffer, while FP represents the number of incorrectly predicted intersections. Similarly, predicted intersection points are used as the center to create another buffer with the same radius R. In this case, tTP represents the number of true intersection points within the buffer, and FN denotes the number of true intersection points not covered by any buffers. In this study, R is set to ten pixels (5 m).

C. Implementation Details

Our algorithm is implemented on an Ubuntu 20 system using the PyTorch framework. The Adam [49] algorithm is utilized for optimization with a learning rate of 0.0001, weight decay factor of 0.0005, $\beta 1$ of 0.9, $\beta 2$ of 0.999, and ε of 1e-8. We utilize two NVIDIA RTX 4090 24 GB GPUs to perform parallel model training. The total number of training epochs is set to 120, but early termination is triggered if the training loss fails to decrease for ten consecutive epochs. During the training phase, we implemented data augmentation techniques, including horizontal flips, vertical flips, rotations, shifting, and scaling. The weight factor ratio in the loss function is $\beta{:}\eta{:}\zeta = 20{:}1{:}1$.

D. Road Surface Segmentation Experiment

In this section, we conducted experiments to compare the performance of the proposed MFFANet with the existing semantic segmentation networks in road extraction. Classic semantic segmentation networks include pyramid scene parsing network (PSPNet) [50], ResUNet [51], U-Net [52], and SegFormer [53]. Methods specifically designed for road extraction include RoadCon [54], LinkNet [55], D-LinkNet [32], CMMPNet [31], FRCFNet [37], MSMDFFNet [56], and OARENet [57]. All results were obtained from the test set and compared using the best model acquired during training.

1) Experiments on the BJRoad Dataset: Table I presents the performance of MFFANet (utilizing the CRSP result) alongside other methods for road semantic segmentation on the BJRoad dataset. In the quantitative comparison, our MFFANet achieved the best results with P_S at 86.89%, R_S at 82.60%,

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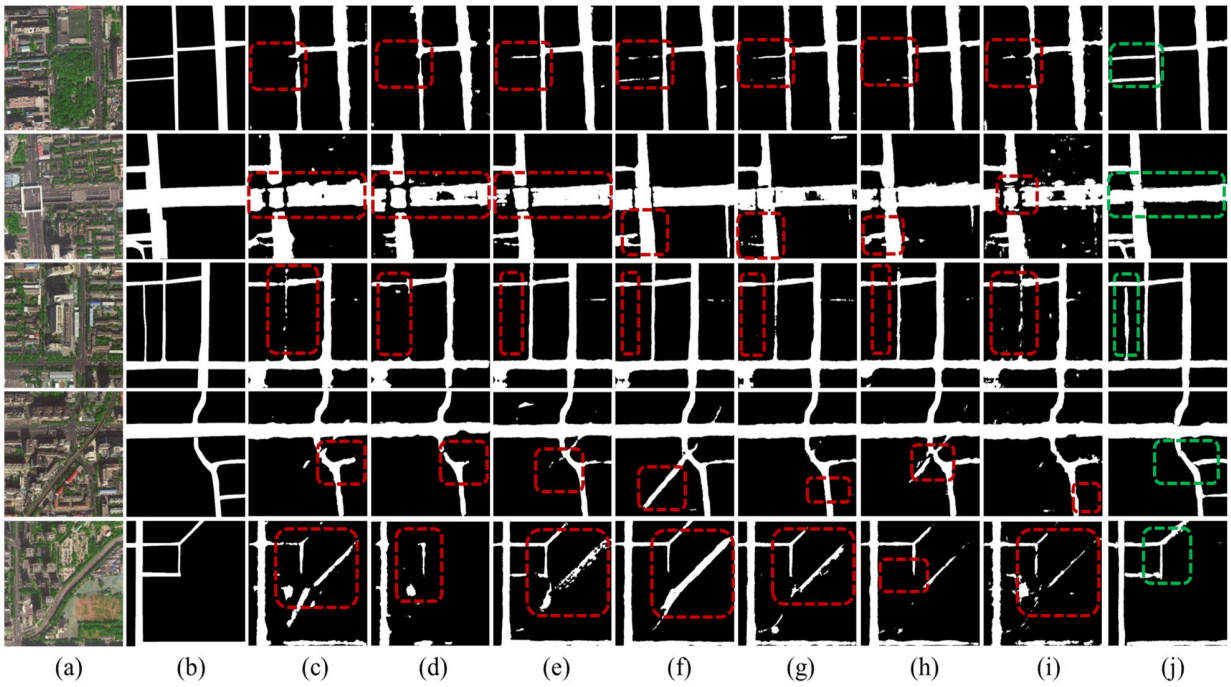


Fig. 4. Segmentation results of different methods on the BJRoad dataset. (a) Images. (b) Label. (c) ResUNet. (d) SegFormer. (e) RoadCon. (f) D-LinkNet. (g) CMMPNet. (h) OARENet. (i) MSMDFFNet. (j) Ours (CRSP).

TABLE I
EXPERIMENTAL RESULTS OF DIFFERENT SEGMENTATION METHODS APPLIED TO THE BJROAD DATASET

Methods	P_S (%)	R_S (%)	$F1_S$ (%)	IoU_S (%)
PSPNet	86.14	68.20	75.18	61.53
ResUnet	80.09	74.97	76.15	62.88
U_Net	77.78	71.33	73.30	59.27
FRCFNet	82.86	69.44	74.35	60.87
SegFormer	81.37	72.45	74.64	61.62
RoadCon	83.39	80.86	81.47	69.40
MSMDFFNet	77.63	76.04	75.70	62.23
LinkNet	83.93	81.02	81.50	69.74
D_LinkNet	83.34	82.79	82.22	70.68
CMMPNet	81.53	**84.30**	82.46	70.43
OARENet	84.08	77.79	79.90	67.72
Ours(CRSP)	**86.89**	82.60	**83.94**	**72.69**

$F1_S$ at 83.94%, and an IoU_S of 72.69%. Our MFFANet outperformed the best ResUNet in classic semantic segmentation networks, with P_S improved by 6.8%, R_S by 7.63%, $F1_S$ by 7.79%, and IoU_S by 9.81%. Furthermore, when compared with specialized road extraction semantic segmentation networks, MFFANet surpassed the best D_LinkNet, achieving a P_S increase of 3.55%, $F1_S$ increase of 1.72%, and IoU_S increase of 2.01%. These results demonstrate the superiority of MFFANet in road extraction through the fusion of multisource data, which can extract roads more comprehensively, overcome the influence of complex environments, and produce segmentation results that are more consistent with ground truth.

We also found that networks specifically designed for road segmentation generally achieve higher metrics compared with classic semantic segmentation networks. We believe that the modules designed for extracting road features play a crucial role. For example, MSMDFF-Net introduces a strip convolution module with a directional parameter; D-LinkNet implements cascaded and parallel dilated convolutions; CMMPNet contains a dual enhancement module for multisource feature interaction; and OARENet designs an SOA module that adaptively fits the spatial details of road regions with dense occlusions through stochastic convolutions. Moreover, our network contains MFFM and MMGM, which play a crucial role in capturing global road information, ensuring the completeness of road extraction, and maintaining topological correctness.

Qualitative comparison of visual results is shown in Fig. 4. It includes results from two classic semantic segmentation networks and five methods specifically designed for road extraction. This further illustrates that the proposed method can extract road masks more accurately in various complex scenarios. The red markers highlight areas where road extraction errors are evident in the comparison methods, while the green areas showcase the accurate extraction achieved by our MFFANet. These visual comparison results are consistent with the quantitative metrics. In the first row, we observe that the roads are heavily obscured by building shadows, making it difficult to detect their locations. This considerably tests the network's feature extraction capability. While other methods fail, our approach effectively captures global features and aggregates contextual information to identify almost all roads accurately despite the occlusion. In the second row, facing interference from pedestrian overpasses, the comparison methods show fragmentation and holes, while our MFFANet can still infer continuous road results. In the third row, the

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TABLE II
EXPERIMENTAL RESULTS OF DIFFERENT SEGMENTATION METHODS APPLIED TO THE PWROAD DATASET

Methods	P_S (%)	R_S (%)	$F1_S$ (%)	IoU_S (%)
PSPNet	**91.91**	67.53	77.09	63.40
ResUnet	84.54	70.74	76.48	62.31
U_Net	88.16	69.77	77.30	63.47
FRCFNet	86.87	67.35	75.16	60.65
MSMDFFNet	84.86	73.03	77.96	64.20
SegFormer	81.56	75.81	78.36	64.53
RoadCon	88.72	73.51	79.79	66.87
OARENet	85.57	78.37	81.38	69.07
LinkNet	88.67	78.82	83.02	71.31
D_LinkNet	89.11	76.82	82.09	69.98
CMMPNet	85.36	84.33	84.57	73.45
Ours (CRSP)	86.96	**88.48**	**87.50**	**77.89**

road is heavily obscured by trees, making it difficult to detect. The comparison methods have poor results in such challenging conditions. However, our model effectively overcomes these environmental obstacles by integrating features from multiple modalities, accurately identifying the roads despite the obstructions. In the fourth and fifth rows, there is a railway and a drainage ditch, respectively, both of which have spectral shapes similar to roads. RoadCon, D-LinkNet, CMMPNet, and OARENet made extraction errors in these cases. However, our network excels at accurately distinguishing the actual roads to be extracted, as MFFANet enhances road feature semantics and effectively suppresses distractions through a multistage feature attention fusion mechanism. According to the analyses above, our model maintains higher accuracy and connectivity in the face of complex scenarios, as MFFANet utilizes MFFM to obtain better contextual information about the roads, and considers the local and global optimality of the roads through MMGM, leading to more comprehensive road extraction results. Overall, the roads segmented by MFFANet exhibit better connectivity and completeness than those from other methods.

2) Experiments on the PWRoad Dataset: Table II presents the segmentation results of our MFFANet and other comparative models. Compared to the BJRoad dataset, most models exhibit an increase in IoU_S, with MFFANet notably achieving a 5.2% improvement. This result can be attributed to the larger number of training samples and more precise road annotations in the PWRoad dataset. Compared with methods such as CMMPNet and D-LinkNet, our MFFANet method exhibits superior performance in comprehensive metrics. Compared with classical semantic segmentation networks, our MFFANet outperforms the best SegFormer by 13.36% in IoU_S. Compared with specialized semantic segmentation networks for road extraction, MFFANet outperforms the best CMMPNet by 4.44% in IoU_S.

Fig. 5 shows some typical road extraction results, with red and green boxes highlighting areas of remarkable improvement. Our method performs better than others in terms of road integrity and connectivity. The first row displays a railway that closely resembles the morphology of a road. In the second and third rows, the roads are partially obstructed by trees and buildings. The last two rows display a densely distributed road network. The results indicate that MFFANet can effectively counteract interference factors and accurately identify roads under these circumstances. Through the fusion of semantic information using the attention mechanism, our model enhances road semantics and maintains road continuity. In contrast, other methods, such as OARENet, CMMPNet, and RoadCon, suffer from serious issues of omission when confronted with occlusions and dense road distributions.

TABLE III
ABLATION EXPERIMENT RESULTS FOR DIFFERENT DATE FUSION ON THE BJROAD DATASET

Settings	P_S (%)	R_S (%)	$F1_S$ (%)	IoU_S (%)
Model-T	76.36	70.02	71.36	57.90
Model-I	80.79	74.41	76.08	62.56
Model-IT	80.89	73.69	75.86	62.57
Model -TO	79.50	80.25	78.82	66.10
Model-IO	85.06	82.09	82.58	70.75
Model-ITO	**86.89**	**82.60**	**83.94**	**72.69**

E. Data Ablation Experiment

In this section, we validate the effectiveness of multisource data fusion in road mask extraction on the BJRoad dataset through ablation experiments. We designed five fusion methods for experimentation: 1) Model-T: extracting road masks using only trajectory data; 2) Model-I: extracting road masks using only image; 3) Model-IT: fusion of image and trajectory for road mask extraction; 4) Model-TO: fusion of trajectory and OSM points for road mask extraction; 5) Model-IO: fusion of image and OSM points for road mask extraction; and 6) Model-ITO: fusion of image, OSM, and trajectory for road mask extraction. The results of the ablation experiments are shown in Table III. Model-T, which only contains trajectory data, performs the worst. Model-ITO, which simultaneously incorporates all three types of data, achieves the best results, with a 10.53% higher P_S, 12.58% higher R_S, 12.58% higher $F1_S$, and 14.79% higher IoU_S compared with Model-T. Model-IO, which includes image and trajectory data, outperforms Model-I, which only contains image data, with improvements of 4.27%, 7.68%, 6.5%, and 8.19% in P_S, R_S, $F1_S$, and IoU_S, respectively. Model-ITO performs better than Model-IO with increases in P_S, R_S, $F1_S$, and IoU_S of 1.83%, 0.51%, 1.36%, and 1.94%, respectively. The quantitative metrics illustrate that fusing features from multiple data sources can effectively enhance road segmentation performance.

Additionally, Fig. 6 presents the segmentation results achieved using various data fusion methods. Road extraction results from a single data source, including Model-T and Model-I, show the worst performance with severe road omissions. The fusion of two data sources can generate more accurate road information. For instance, compared with Model-I, both Model-IT and Model-IO, which incorporate trajectory data and OSM points respectively, effectively address the challenges of occluded roads and minimize interference

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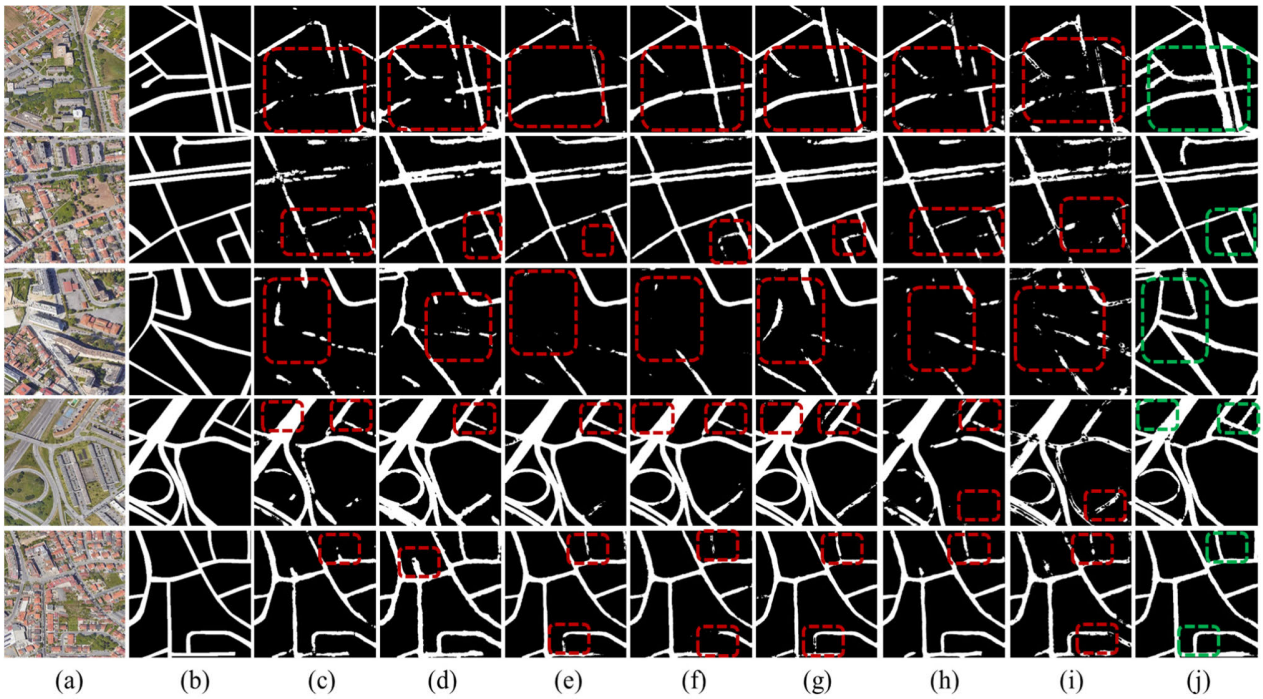


Fig. 5. Segmentation results of different methods on the PWRoad dataset. (a) Images. (b) Label. (c) ResUNet. (d) SegFormer. (e) RoadCon. (f) D-LinkNet. (g) CMMPNet. (h) OARENet. (i) MSMDFFNet. (j) Ours (CRSP).

from ditches and railroads, enabling them to infer the correct locations of roads. The segmentation map produced by Model-ITO, which integrates all three data sources, closely resembles the ground truth, providing a more comprehensive road extraction compared with the fusion of only two data sources. In summary, single data source inevitably has limitations because roads exhibit diverse morphologies and are often located within complex backgrounds. Therefore, leveraging the complementary features from multisource data fusion is undoubtedly the most effective method for road extraction.

F. Multilevel Road Mask Fusion

In our proposed method, the quality of extracting edges, centerlines, and intersections is highly dependent on the road masks. To obtain high-quality masks, we meticulously designed three strategies: 1) multilevel road mask generation, including LRSP, CRSP, and CRNP; 2) fusion of multilevel mask information; and 3) regularization of road mask generation. In this section, we validate the effectiveness of these strategies through experiments conducted on BJRoad dataset, as described in Table IV. In terms of IoU_S, LRSP is 1.19% and 0.38% higher than CRSP and CRNP, respectively. Moreover, LRSP has better R_S, approximately 10% higher than CRSP and CRNP. CRSP and CRNP exhibit better P_S, roughly 8% higher than LRSP. The fusion results show IoU_S improvements of 0.14%, 1.33%, and 0.52% over LRSP, CRSP, and CRNP, respectively. $F1_S$ and IoU_S of the regularized masks are lower than the fusion results by 0.4% and 0.56%, respectively. IoU_S of LRSP is higher than that of CRSP and CRNP because the network generates multiple predictions for the same road section at different locations, enabling it to extract more challenging roads effectively. The above results show that our carefully designed strategies for various mask processings are meaningful. Our strategies not only enhance the completeness and reliability of road extraction but also outperform the best current methods in comprehensive metrics, ensuring the quality of subsequent extractions of centerline, edge, and intersection.

TABLE IV
MULTILEVEL RESULTS, FUSION RESULTS, AND REGULARIZATION RESULTS ON THE BJROAD DATASET

Types	P_S (%)	R_S (%)	$F1_S$ (%)	IoU_S (%)
LRSP	78.51	**93.16**	84.54	73.88
CRSP	86.89	82.60	83.94	72.69
CRNP	86.76	83.57	84.48	73.50
Fusion	78.80	93.10	**84.75**	**74.02**
Regulation	**88.39**	82.00	84.35	73.46

In Fig. 7, we separately combined the results of LRSP, CRSP, and CRNP. As shown in Fig. 7(f), the road quality improves after fusion, effectively eliminating holes and noise points in the roads. Compared with the fused results, the quality of the regularized road masks is further enhanced, particularly in improving the correctness of road topology and better addressing the issues related to road jaggedness and discontinuity. As shown in Fig. 6(g), the road edges are smooth, with enhanced details, and more complete road connections.

G. Regularization Experiment

In this section, we first need to specify several key parameter values. For noise removal, we set the area threshold S to 50 m^2,

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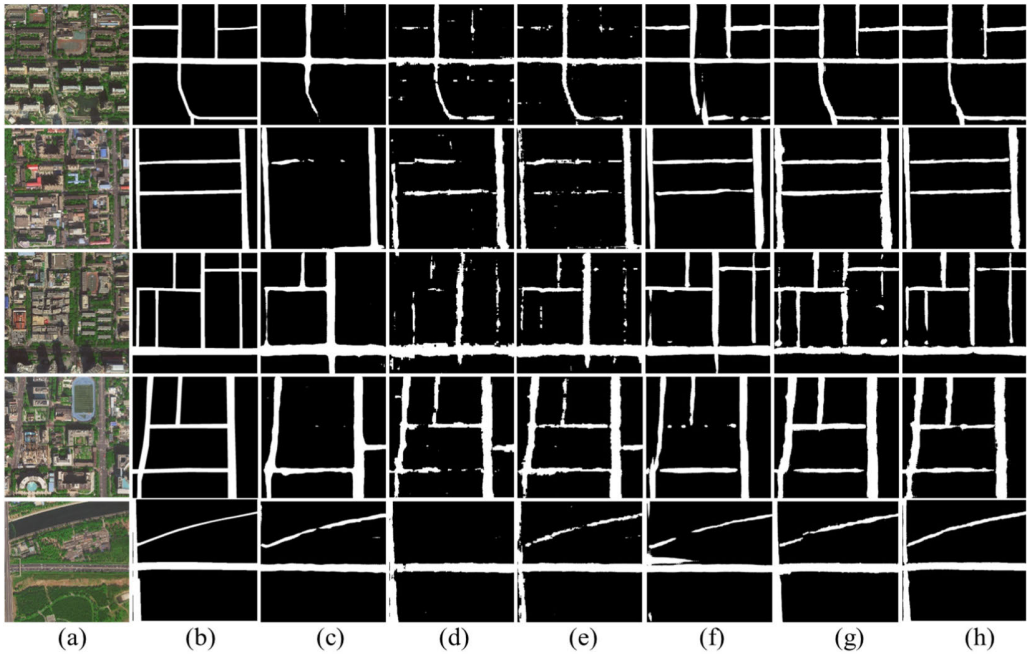


Fig. 6. Segmentation results of different data fusion on the BJRoad dataset. (a) Images. (b) Label. (c) Model-T. (d) Model-I. (e) Model-IT. (f) Model-TO. (g) Model-IO. (h) Model-ITO.

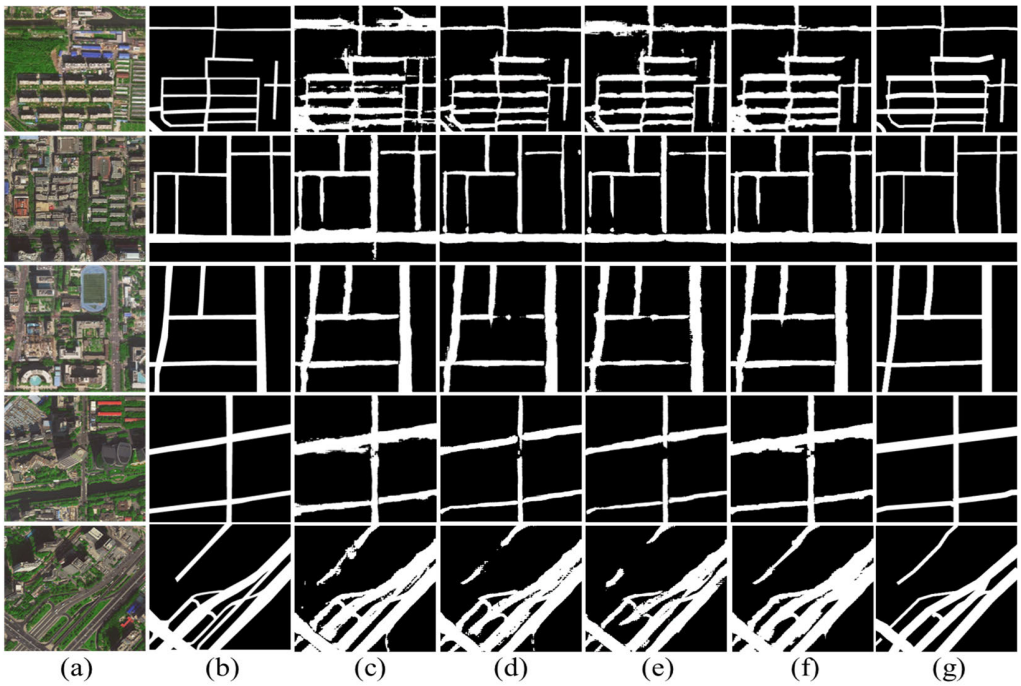


Fig. 7. Segmentation results of multilevel masks, fusion, and regularization on the BJRoad dataset. (a) Images. (b) Label. (c) LRSP. (d) CRSP. (e) CRNP. (f) Fusion. (g) Regularization.

treating road patches smaller than this threshold as potential noise to be eliminated. Another important parameter, ε, which is used to determine whether to retain or remove larger noise patches, is highly correlated with α and β, and is set to $\pi/26$. Furthermore, in coarse centerline spikes localization, we define four groups of minimum rectangular aspect ratio ranges, namely, [0, 0.1), [0.1, 0.2), [0.2, 0.32), and [0.32, $+\infty$), and provide different angle judgment conditions to localize spikes in each range. The tolerance distance in the DP algorithm is set in accordance with the area of road patches. This parameter aims to remove redundant points in the centerline, restoring the straight and smooth features of the road. Although some parameters seem to be empirically set, the following experiments on different datasets all demonstrate that our regularization algorithm is robust under these fixed threshold settings.

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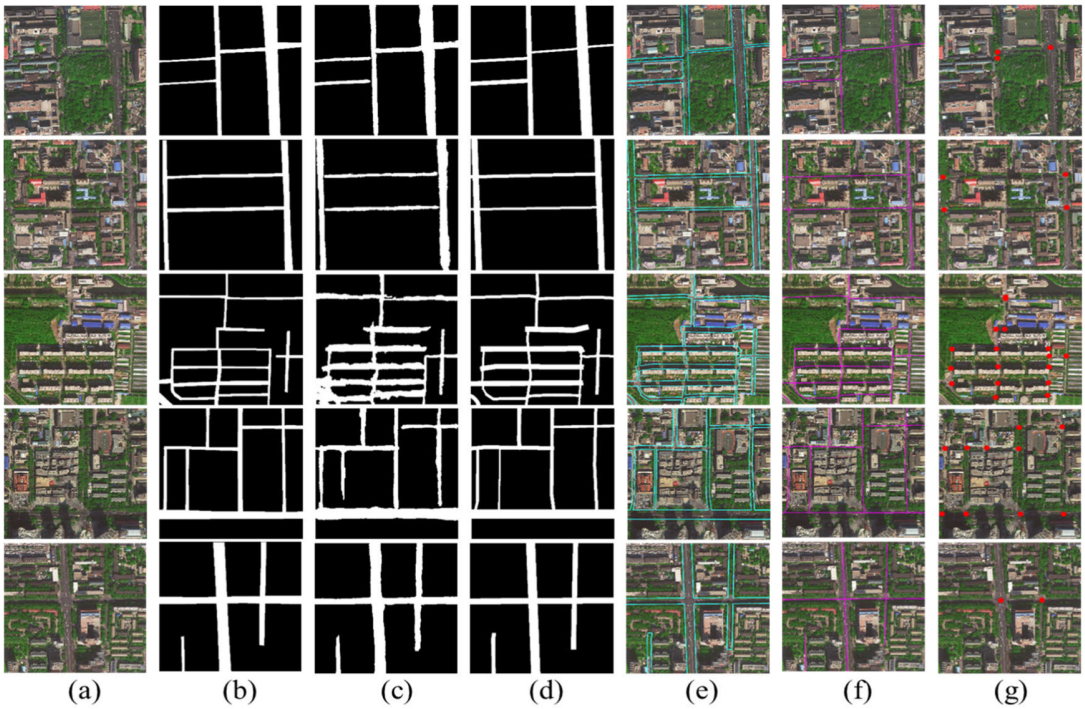


Fig. 8. Road regularization results on the BJRoad dataset. (a) Images. (b) Label. (c) Multilevel mask fusion results. (d) Regularization masks. Images with (e) road edge lines, (f) road centerlines, and (g) intersections.

1) Experiments on the BJRoad Dataset: Fig. 8 illustrates five regularization results on the BJRoad dataset, including multilevel mask fusion results, regularized mask results, road edge line results depicted in cyan blue, road centerline results shown in magenta, and intersection point information marked in red. In the second and fourth rows, given the occlusion by trees and shadows from buildings, the road discontinuity issue in the fused mask results remains but is resolved in the regularized results by adjusting the relationships between road segments. The edge line road network and centerline road network have correct topological connections, consistent with the actual road network. The road in the third row of Fig. 8 shows a common road type in the area. In Fig. 8(c), an error is visible in the bottom-left corner which is typically due to the imperfect segmentation and fusion. However, it is effectively corrected by our regularization algorithm. The corresponding edge line and centerline road networks exhibit good positional accuracy. Observing all examples in the third column reveals a common problem that may exist in almost all road semantic segmentation methods, that is, the width of the same road segment varies. However, our mask regularization strategy effectively addresses this problem, producing results that are much closer to the ground truth. Finally, through the regularization algorithm, we obtain nearly perfect road bilateral line networks and centerline networks, as shown in columns five and six, respectively. Additionally, we consider intersections, which are highlighted in column seven.

Table V presents the quantitative descriptive results of road edge line, centerline, and intersection extraction after road surface vectorization. On the BJRoad dataset, the P_L of road edge line extraction is 74.09%, R_L is 70.19%, and $F1_L$ is 72.09%. For the road centerline extraction, the P_L, R_L, and $F1_L$ are 79.96%, 78.32%, and 79.13%, respectively. The P_I of intersection extraction is 76.17%, with an R_I of 71.36% and an $F1_I$ of 73.68%. These results further demonstrate the reliability of our method in extracting road edge lines, centerlines, and intersections.

TABLE V
QUANTITATIVE EXPERIMENTAL RESULTS OF ROAD EDGE LINE, CENTERLINE, AND INTERSECTION EXTRACTION ON THE BJROAD DATASET

Types	Precision(%)	Recall(%)	F1(%)
Road Edge Lines	74.09	70.19	72.09
Road Centerlines	79.96	78.32	79.13
Intersections	76.17	71.36	73.68

2) Experiments on the PWRoad Dataset: Fig. 9 presents several typical examples processed with regularization on the PWRoad dataset. It includes multilevel mask fusion results, regularized mask results, road edge line results depicted in cyan blue, road centerline results shown in magenta, and intersection point information marked in red. Our method continues to perform well when faced with densely distributed roads. The regularized masks exhibit superior connectivity and edge details compared with fused masks. The extracted edge line road network and centerline road network have both well topological connectivity and positional accuracy. In addition, intersection locations can be effectively extracted, as shown in Fig. 9(g). Therefore, our regularization algorithm is robust to different datasets and different road styles.

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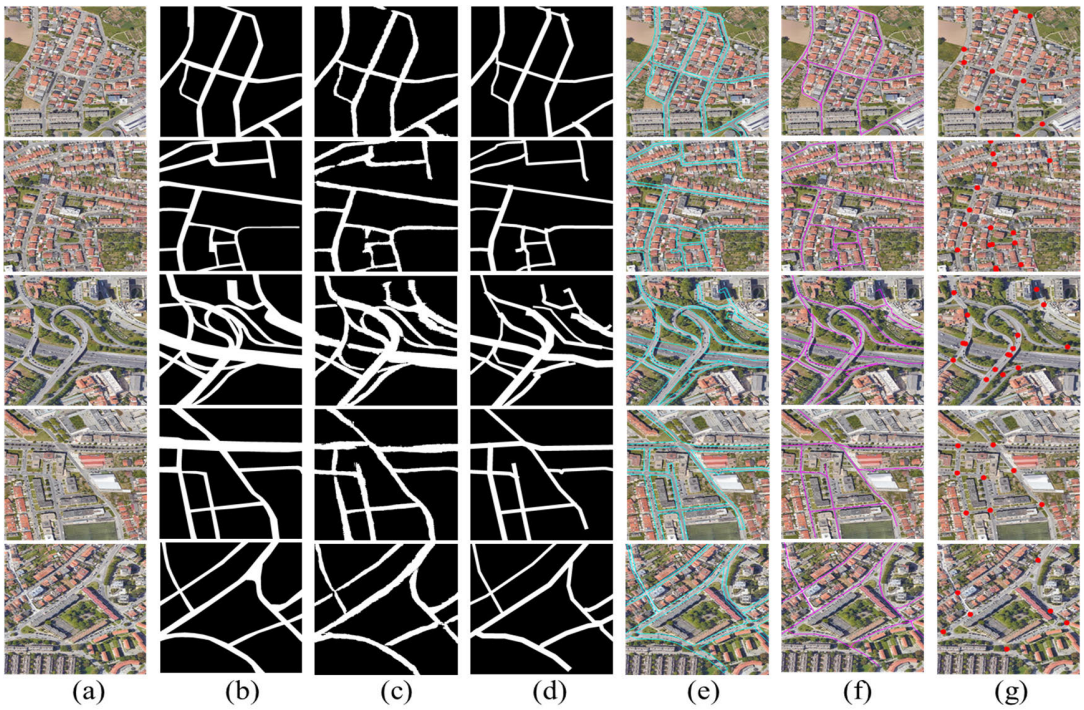


Fig. 9. Road regularization results on the PWRoad dataset. (a) Images. (b) Label. (c) Multilevel mask fusion results. (d) Regularization masks. Images with (e) road edge lines, (f) road centerlines, and (g) intersections.

TABLE VI
QUANTITATIVE EXPERIMENTAL RESULTS OF ROAD EDGE LINE, CENTERLINE, AND INTERSECTION EXTRACTION ON THE PWROAD DATASET

Types	Precision(%)	Recall(%)	F1(%)
Road Edge Lines	78.66	75.23	76.90
Road Centerlines	82.85	84.12	83.48
Intersections	69.57	73.73	71.59

Table VI presents the quantitative evaluation results of road edge line, road centerline, and intersection extraction after road surface vectorization. The metrics of the centerline network and edge network are improved in comparison with Table V. Specifically, the P_L, R_L, and $F1_L$ of the edge line network are 78.66%, 75.23%, and 76.90%, respectively. For the centerline network, the P_L, R_L, and $F1_L$ are 82.85%, 84.12%, and 83.48%, respectively. It further demonstrates that the designed algorithm is effective in solving the road surface vectorization problem.

Based on the experimental results, our regularization algorithm performs exceptionally well in validating its effectiveness and solving the road surface vectorization problem. Through regularization, the connectivity and edge details of the road mask are significantly improved, effectively avoiding the road jaggies and discontinuities. The experimental results in Figs. 8 and 9 show that the regularized mask outperforms the original road surface segmentation results in terms of connectivity and detail recovery, especially in densely distributed road scenes, where the regularization algorithm better preserves the road topology. At the same time, the regularization algorithm effectively addresses the road surface vectorization problem, resulting in the extraction of road centerlines, edge lines, and intersections. Figs. 8 and 9 demonstrate that the extracted road centerlines and edge lines are accurately positioned, maintaining the road topology structure. The results in Tables V and VI show that the comprehensive metrics of road vectorization exceed 70%, highlighting the advantages of the proposed algorithm in multitask extraction.

H. Centerline Extraction Comparison Experiment

Our method is compared with two state-of-the-art graph-based baseline methods. Qualitative evaluations on two datasets are presented in Figs. 10 and 11, respectively. Each image has a size of 1024 × 1024 pixels. Specifically, Figs. 10(a) and 11(a) show the ground-truth road network (cyan blue lines), and Figs. 10(b)–(d) and 11(b)–(d) are road networks (magenta lines) predicted from different models, where (b) is RNGDet++, (c) is SAM-Road, and (d) is our proposed method. For enhanced visualization, the lines are slightly widened, but the width is only one pixel in reality. According to these results, it is evident that RNGDet++ has a major issue of missing roads. Compared with RNGDet++, SAM-Road clearly predicts more roads; however, it suffers from issues such as redundant roads at intersections and some segments appearing as wavy lines, which are inconsistent with reality. Our method effectively overcomes the issues present in RNGDet++ and SAM-Road, with a complete road structure and correct topological connections. Even in highly challenging situations, such as dense urban areas with many blocks and intersections, or irregularly curved roads, our method can consistently predict road networks with high accuracy.

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Fig. 10. Road centerline results of different methods on the BJRoad dataset. (a) Ground truth. (b) RNGDet++. (c) SAM-Road. (d) Ours.



Fig. 11. Road centerline results of different methods on the PWRoad dataset. (a) Ground truth. (b) RNGDet++. (c) SAM-Road. (d) Ours.

TABLE VII
EXPERIMENTAL RESULTS OF DIFFERENT CENTERLINE EXTRACTION METHODS ON THE BJROAD DATASET

Methods	P_L (%)	R_L (%)	$F1_L$ (%)
RNGDet++	79.17	70.67	74.68
SAM-Road	**81.84**	73.05	77.20
Ours	79.96	**78.32**	**79.13**

TABLE VIII
EXPERIMENTAL RESULTS OF DIFFERENT CENTERLINE EXTRACTION METHODS ON THE PWROAD DATASET

Methods	P_L (%)	R_L (%)	$F1_L$ (%)
RNGDet++	81.19	77.89	79.50
SAM-Road	**83.01**	79.63	81.28
Ours	82.85	**84.12**	**83.48**

The quantitative evaluation results on the BJRoad dataset and PWRoad dataset are shown in Tables VII and VIII, respectively. The highest scores are highlighted in bold. Our method achieves better scores on the metrics of R_L and $F1_L$, while SAM-Road has the highest P_L. The R_L of our method is 7.65% and 5.27% higher than that of RNGDet++ and SAM-Road on the BJRoad dataset and 6.23% and 4.49% higher on the PWRoad dataset, respectively. This result indicates that our method can predict road centerlines more comprehensively, while RNGDet++ and SAM-Road have issues about missing road centerlines.

The outstanding performance of our method can be largely attributed to the design of the multilevel road mask output network and the refined regularization algorithm. The masks predicted by our multisource data-based network are sharp and clear, thus ensuring the integrity of road extraction. Additionally, the refined postprocessing algorithm makes full use of spatial and geometric information based on existing road rules, which guarantees the correct topology.

IV. DISCUSSION

A. Strategies for Extracting Complete and Continuous Roads

Many studies have focused on extracting roads from remote sensing images, but a common issue in these studies is road fragmentation and disconnection. The reasons for these problems are as follows.

1) Remote sensing images have certain limitations, such as roads often being obscured by shadows from buildings and trees and road textures being similar to those of other objects, which make it difficult to detect roads based on images.
2) The capability of the network to extract roads is another key factor. General networks often struggle to capture the characteristics of long-distance roads, leading to considerable omissions and misextractions. In contrast, networks that account for road-specific characteristics show superior road extraction performance, as demonstrated in Figs. 4 and 5.

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3) The design of the loss function significantly affects the model's performance. The road extraction method proposed in this study effectively addresses the aforementioned challenges, providing an improved solution to the issues of disconnection and fragmentation when extracting roads from images, resulting in smooth and complete road segmentation results. We incorporated vehicle trajectories and OSM points for road extraction, which compensated for the obstacles and texture interference in the images. The results presented in Table III and Fig. 6 clearly demonstrate the importance of multisource data fusion in enhancing road extraction.

To capture road characteristics, we designed the CFEM, MFFM, MMGM, and a joint loss function. These enhancements enable our model to outperform other networks significantly in road extraction. As shown in Tables I and II, the IoU_S of our model is 2.01% and 4.44% higher than the second-best model, respectively. As shown in Figs. 4 and 5, the roads extracted using our method are more complete and more continuous. This result demonstrates that our approach, which considers road characteristics, is effective and provides a reliable solution for extracting roads with enhanced completeness and continuity.

B. Influence of Road Segmentation on Centerline, Edge Line, and Intersection Extraction

In previous works, pixel-level segmentation of roads has been widely studied, while road vectorization has received comparatively less attention, lacking refined processing of pixel-level roads. To address these issues, this study refines multilevel road masks based on the existing road rules to obtain high-quality road centerlines, edge lines, and intersections. Performance comparison of our method on the BJRoad and PWRoad datasets reveals that improvements in road semantic segmentation quality lead to high metric scores of vector extraction. Comparing Tables I and II shows that as the IoU_S of our method increases from 72.69% to 77.89%, the corresponding centerline $F1_L$ improves by 4.35%, and the edge line $F1_L$ increases by 4.81%. This result illustrates that the quality of road segmentation significantly impacts the results of vectorization. Fortunately, by introducing multisource data, multistage feature fusion, multilevel mask generation, and joint loss strategy, we can ensure the quality of road segmentation. Previous segmentation methods produced a single road network mask output, which limits postprocessing flexibility and results in significant accuracy loss. In contrast, our regularization algorithm enhances the completeness and reliability of the extraction of centerlines, edge lines, and intersections by considering both local road segments and the global road network.

V. Conclusion

In this study, we proposed a method that utilizes MFFANet and the empirical regularization algorithm to address the challenges of road discontinuity and vectorization in road extraction from high-resolution remote sensing imagery. In MFFANet, we designed the MFFM to efficiently acquire the fusion of complementary features from different modalities, which can capture long-range spatial contextual information and alleviate local interference problems. In addition, MMGM was proposed for simultaneously extracting local road segments, road sections, and road network masks, which is helpful to improve the completeness and continuity of roads. Our proposed MFFANet effectively addresses the issue of road discontinuity and is highly applicable to various road extraction tasks. Moreover, we have dedicated significant efforts to designing an empirical regularization algorithm that refines multistage road masks, resulting in high-quality road centerline networks, edge line networks, and intersections, which greatly expands the applications of road extraction tasks. In this study, we successfully derived the centerlines, edge lines, and intersections through elaborate processing of segmentation results. However, our vectorization process depends on the quality of road surface segmentation. In future work, we will design an end-to-end network that can directly extract road surfaces, centerlines, edge lines, and intersections, offering a more efficient solution for extensive applications of road data.

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